

Fig. 5: Telegram bot app ID

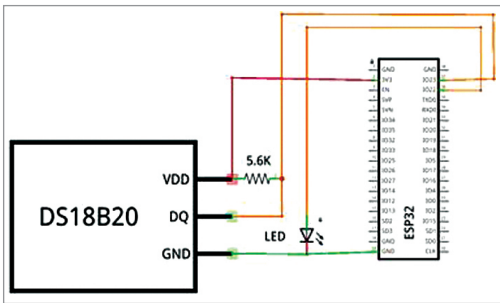


Fig. 6: Circuit diagram of the Telegram with Python

with the temperature added at the last, such as 08/09/2021, 06:35:42, 0746, 0876, 3278, 3563, 2795, 3012

The last data (3012) is the temperature (30.12°C). Hold the DS18B20 in your palm and you would notice that the value is changing due to the heat of your palm. The whole string is uploaded to the Telegram channel by using the simple GET statement of ESP32:

[https://api.telegram.org/bot\\$token/sendMessage?chat_id=\\$chat&text=Hello+World+how+are+you](https://api.telegram.org/bot$token/sendMessage?chat_id=$chat&text=Hello+World+how+are+you)

The same can be achieved using a computer as well, but the ESP32 used is a continuous, small, automatic, and a real hardware for delivering data to Telegram (without a bulky computer) and using just a fraction of the power of a 3.3V source.

(The author has suggested this simple version of the project that can be deployed easily, but the actual project made by him is much more sophisticated. It has a remote sender unit, which sends field sensor data over a LoRa network to another ESP32, which in turn pumps data to the Telegram channel using LoRa on one side and Wi-Fi on the other side—just like a router.)

Receiver unit

This is the crux of the project. You can receive highly secured and encrypted data on your phone. The first test data reaches your mobile only. But you can hardly process that data to do something meaningful, like putting it on a web database for trending, run a relay, or to run a system to perform differently depending on the value of data.

Therefore, replace the mobile phone with an internet connected computer with Python3 installed in it. The Python script will gather data and then re-display it on

the computer at the same speed at which it is being sent at the sending end. Since all data is available as variables inside Python, one can process it any desired way.

While IoT analytics platforms like ThingSpeak can give you a maximum of four data strings per minute, in Telegram you can pump as many as thirty data strings per minute, and yet all are highly secured and encrypted. Below is a data string that arrives every two seconds and it is received in the Python Telegram terminal at the other end of the globe with not a single stream lost:

```
08/09/2021 08:22:15 0746 3563
0876 2795 3278 2962
08/09/2021 08:22:16 0746 3563
0876 2795 3278 2962
08/09/2021 08:22:18 0746 3563
0876 2795 3278 2962
08/09/2021 08:22:20 0746 3563
0876 2795 3278 2962
08/09/2021 08:22:21 0746 3563
0876 2795 3278 2968
08/09/2021 08:22:23 0746 3563
```

```
0876 2795 3278 2968
08/09/2021 08:22:25 0746 3563
0876 2795 3278 2968
08/09/2021 08:22:26 0746 3563
0876 2795 3278 2968
08/09/2021 08:22:28 0746 3563
0876 2795 3278 2962
08/09/2021 08:22:30 0746 3563
0876 2795 3278 2962
```

Software *arduino_sketch.ino* and *python_script.py* can be download from the link <https://www.electronicsforu.com/electronics-projects/hardware-diy/published-telegram-received-python>

Testing

After connecting ESP32 with the sensor and powering the board, and making sure ESP32 and receiver system both have net connectivity, run the Python file at receiver end. You will be able to see the sensor message in Python terminal. You can also get the live sensor data in Telegram as chat.

Once you have successfully learnt to run the sample project, you will be able to deploy Telegram for a real use. On the other end, after receiving it on Python3, you can further process it to run your devices or dump it in a database. With fibre-to-home internet connection, the author could get thirty readings per minute (one 49-character long data reading every two seconds), which is extremely good. If the network slows down, the number of uploads will of course be reduced.

EFY note. One can increase the throughput 'once per second' even but that means immense trafficking in the telegram highway—which is deprecated by the telegram and they will make you debarred in no time. Therefore, never try to increase the data trafficking less than a two second period. **EFY**

EFY Note

The source code of this project is available for free download at source.efymag.com



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